Urban Green Space and Urban Heat: A Multi-City Spatial Analysis of NDVI and Land Surface Temperature in the United States

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Abstract

This study investigates the relationship between urban greenness, as measured by the Normalized Difference Vegetation Index (NDVI), and land surface temperature (LST) across ten major U.S. cities. Using satellite-derived NDVI and LST data combined with U.S. Census income data at the tract level, we analyze spatial inequalities in urban heat exposure and vegetation distribution. The results show a strong negative correlation between NDVI and LST in most cities, confirming that greener neighborhoods are significantly cooler. Furthermore, income–NDVI analysis reveals inequitable distribution of green space in several cities, with lower-income neighborhoods tending to have less vegetation. These findings highlight the importance of urban greening strategies in climate adaptation and social equity.

1. Introduction

Climate change and rapid urbanization are driving increasing concern about extreme heat in cities worldwide. The urban heat island (UHI) effect describes the tendency of urban areas to experience higher temperatures than surrounding rural areas, due to the heat-absorbing properties of built environments and the relative lack of vegetation. This effect exacerbates the risks of heat-related illness and mortality, particularly among vulnerable populations.

Urban vegetation, including tree canopy and parks, plays a critical role in mitigating UHIs. Vegetation cools surface temperatures through evapotranspiration and shading. Satellite-derived indicators such as the Normalized Difference Vegetation Index (NDVI) enable large-scale measurement of vegetation cover and its relationship with surface temperature, typically measured through land surface temperature (LST) products.

However, access to green space is not evenly distributed within cities. Numerous studies suggest that lower-income and minority neighborhoods often have less green space and are more exposed to heat. Addressing these inequities is a key goal of climate justice and urban resilience planning.

This project addresses three key questions:

1. How does NDVI relate to LST at the neighborhood (tract) level across major U.S. cities?
2. How does the strength of this relationship vary between cities with different climates?
3. Is green space distributed equitably with respect to neighborhood income?

We perform a multi-city analysis using satellite data and U.S. Census data at the tract level for ten major U.S. cities. Our findings provide insight into both environmental and social dimensions of urban climate adaptation.

2. Data and Methods

2.1 Data Sources

The analysis combines four primary data sources:

- NDVI: Normalized Difference Vegetation Index data were derived from MODIS MOD13Q1 imagery for July 2022, with a spatial resolution of 250 meters. NDVI is a widely used measure of vegetation greenness and health.
- LST: Land Surface Temperature was obtained from MODIS MOD11A2 imagery for July 2022, with a spatial resolution of 1 km. LST provides an estimate of surface temperature, which correlates with the UHI effect.
- Census Tracts: The 2020 TIGER/Line shapefiles were used to define neighborhood boundaries at the census tract level. Census tracts provide a consistent unit for spatial analysis of demographic and environmental variables.
- Income: Median household income data were obtained from the 2020 American Community Survey (ACS) 5-year estimates (variable B19013_001).

2.2 Study Area

The study focuses on ten major U.S. cities selected for geographic diversity and population size: New York, Los Angeles, Chicago, Houston, Phoenix, Philadelphia, San Antonio, San Diego, Dallas, and San Jose. These cities represent a range of climate zones, from hot and dry (Phoenix) to coastal temperate (San Diego) and humid continental (Chicago).

2.3 Data Processing

NDVI and LST rasters were downloaded from Google Earth Engine as GeoTIFF files. For each city, the rasters were masked to the extent of the corresponding census tracts using the sf and terra R packages. Mean NDVI and LST values were calculated for each tract using zonal statistics (exactextractr package). This produced a dataset of mean greenness and temperature per neighborhood.
Median household income was obtained using the tidycensus package and joined to the tract-level NDVI and LST dataset. Due to ACS data limitations, income data were incomplete for some cities and tracts; the Income–NDVI analysis was therefore limited to six cities with sufficient data.

2.4 Analytical Approach

NDVI–LST Relationship:
For each city, we computed the correlation between NDVI and LST and fitted a simple linear regression model to estimate the cooling effect of vegetation. We compared results across cities to assess geographic variation.

Income–NDVI Relationship:
In cities with sufficient income data, we analyzed the correlation and regression relationship between income and NDVI to assess spatial equity in green space distribution.

Decile Analysis:
To provide additional insight, we analyzed mean LST across NDVI deciles within each city. This allows us to quantify the temperature difference between the least and most green neighborhoods.

3. Results

3.1 NDVI–LST Relationship

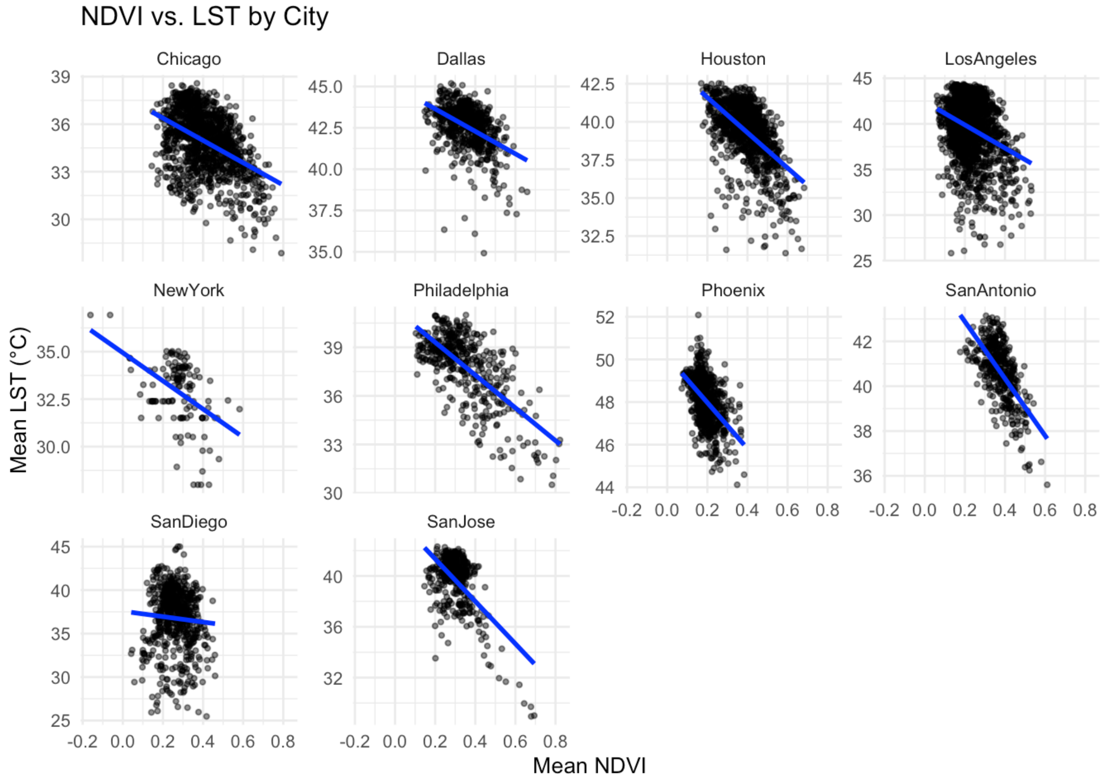
Table 1 summarizes the NDVI–LST relationships across all ten cities.

Table 1. Summary of NDVI–LST Relationship by City

City	Correlation	Slope (°C/NDVI)	R^2	Number of Tracts
Philadelphia	-0.688	-10.12	0.473	408
SanAntonio	-0.630	-12.73	0.397	375
SanJose	-0.582	-16.69	0.338	363
Houston	-0.553	-11.58	0.306	1098
Phoenix	-0.490	-10.80	0.240	730
NewYork	-0.459	-7.41	0.211	124
Dallas	-0.448	-6.80	0.201	600
Chicago	-0.444	-7.05	0.197	1167
LosAngeles	-0.308	-12.70	0.095	1769
SanDiego	-0.062	-3.12	0.004	593

The results demonstrate a consistent negative relationship between greenness and surface temperature. The strength of this relationship varies by city and climate. Philadelphia, San Antonio, and San Jose exhibit the strongest effects, with R^2 values above 0.3 and cooling effects of 10–17°C per NDVI unit. Inland cities such as Phoenix, Houston, and Dallas also show substantial effects. In contrast, coastal cities like San Diego and Los Angeles display weaker relationships, likely due to ocean-moderated temperatures and relatively homogeneous vegetation cover.

Figure 1. NDVI vs LST Scatterplots per City

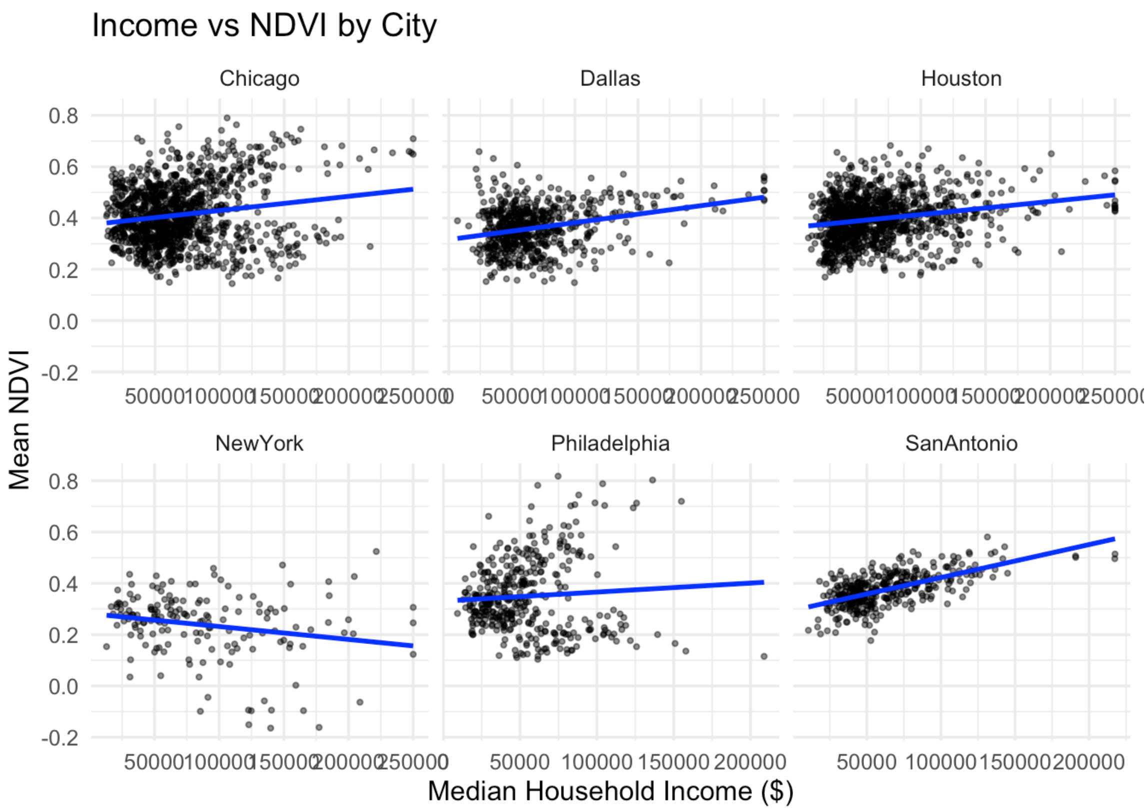


Visual inspection of Figure 1 supports these findings. Cities with strong NDVI–LST relationships display steep negative slopes, while coastal cities show flatter trends.

3.2 Income–NDVI Relationship

Income–NDVI analysis was conducted for six cities with sufficient data (Philadelphia, San Antonio, San Jose, Houston, Phoenix, and Dallas). The results suggest that green space is not distributed equitably across income groups.

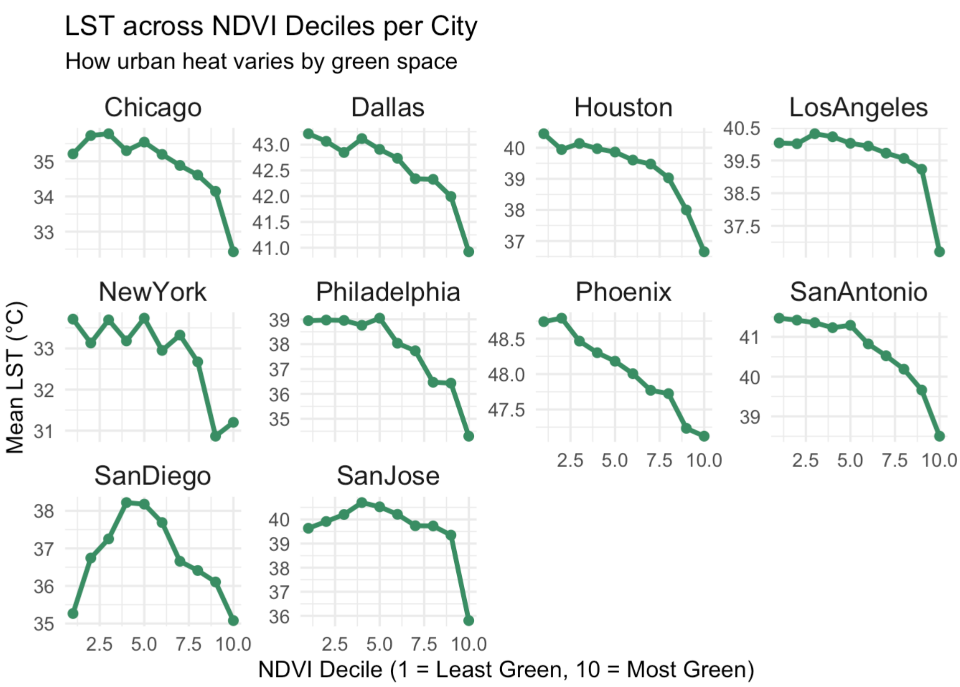
Figure 2. Income vs NDVI Scatterplots per City (6 Cities)



Philadelphia and San Jose exhibit clear positive relationships between income and NDVI, with higher-income neighborhoods tending to be greener. San Antonio and Phoenix display similar trends, though with more variability. Chicago and Dallas show a weaker or more complex relationship.

3.3 Decile Analysis

Figure 3. LST across NDVI Deciles per City



The decile analysis further illustrates the relationship between greenness and urban heat. In most cities, the least green neighborhoods (bottom NDVI decile) are significantly hotter than the greenest neighborhoods (top decile). The temperature difference between bottom and top deciles ranges from 4°C to over 7°C in the strongest cases. This highlights the potential for urban greening interventions to mitigate extreme heat.

4. Discussion

The results of this study confirm that vegetation is an important factor in reducing urban surface temperatures. Across all cities analyzed, greener neighborhoods were cooler, though the strength of this relationship varied by climate zone.
In hot inland cities, where vegetation scarcity and high temperatures coincide, increasing green cover could substantially reduce heat exposure. In coastal cities, where the ocean moderate's temperature and vegetation patterns are more homogeneous, the impact of greenness on LST is less pronounced.
The analysis of income and greenness revealed significant equity concerns. In many cities, lower-income neighborhoods have both less vegetation and higher temperatures, compounding social and environmental vulnerabilities. This finding aligns with prior research on environmental justice and suggests that urban greening policies must explicitly address spatial inequities.

Limitations:

This analysis used data from a single summer month, limiting temporal generalizability. ACS income data were incomplete for some cities, reducing the scope of the Income–NDVI analysis. Additionally, only simple linear models were applied; future work could explore non-linear or spatial models to capture more complex relationships.

5. Conclusion

This study demonstrates that increasing urban greenness can substantially mitigate urban heat at the neighborhood level. However, the benefits of vegetation are not equitably distributed. Lower-income neighborhoods often lack green space and face greater heat exposure, compounding health and social risks.